

1. Business Understanding

- Perform EDA on the data: Univariate, bivariate and multivariate analysis
- Add the model analysis
- following the CRISP-DM methodology
- Undertake exercise on *streamlit*

2. Data Understanding

Upload Diabetes Dataset (.csv)

 Upload

200MB per file • CSV

Dataset Preview

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Pedigree	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Dataset Shape

Rows: 768, Columns: 9

Columns in Dataset

```
▼ [  
  0 : "Pregnancies"  
  1 : "Glucose"  
  2 : "BloodPressure"  
  3 : "SkinThickness"  
  4 : "Insulin"  
  5 : "BMI"  
  6 : "Pedigree"  
  7 : "Age"  
  8 : "Outcome"  
]
```

- The dataset contains multiple medical predictor variables such as Glucose, BMI, Age, and Insulin, alongside the target variable Outcome, which indicates whether a person has diabetes.
- The structure confirms a supervised classification problem with labeled data.

Missing Values

	0
Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
Pedigree	0
Age	0
Outcome	0

Summary Statistics

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Pedigree	Age	Outcome
count	768	768	768	768	768	768	768	768	
mean	3.8451	120.8945	69.1055	20.5365	79.7995	31.9926	0.4719	33.2409	0
std	3.3696	31.9726	19.3558	15.9522	115.244	7.8842	0.3313	11.7602	0
min	0	0	0	0	0	0	0.078	21	
25%	1	99	62	0	0	27.3	0.2438	24	
50%	3	117	72	23	30.5	32	0.3725	29	
75%	6	140.25	80	32	127.25	36.6	0.6263	41	
max	17	199	122	99	846	67.1	2.42	81	

Dataset Information

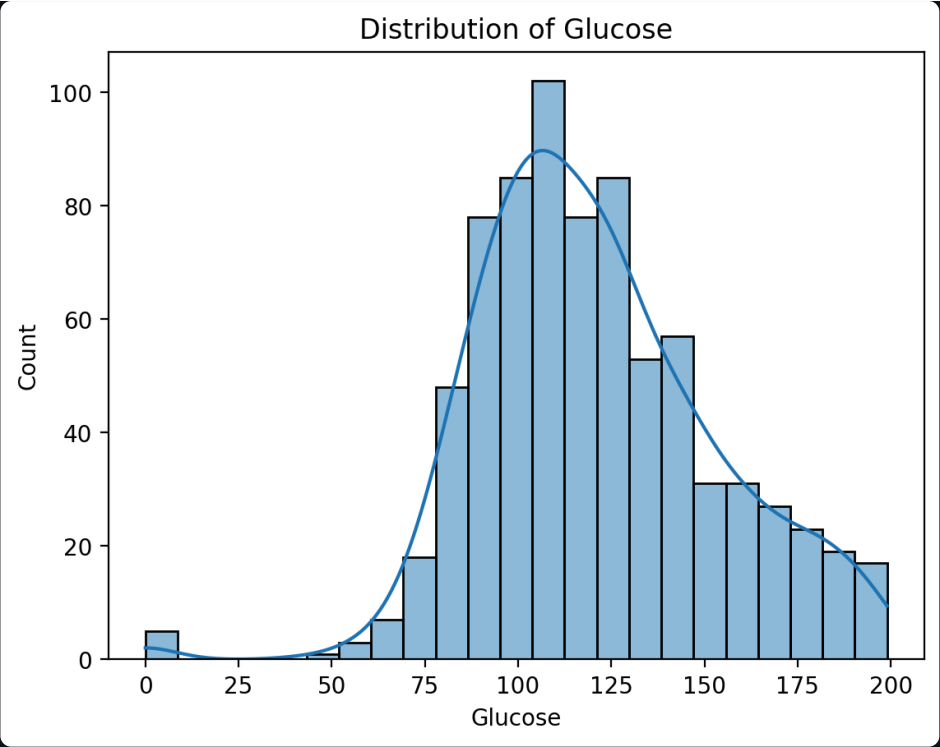
```
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#  Column      Non-Null Count  Dtype
---  -
0  Pregnancies  768 non-null   int64
1  Glucose      768 non-null   int64
2  BloodPressure 768 non-null   int64
3  SkinThickness 768 non-null   int64
4  Insulin      768 non-null   int64
5  BMI          768 non-null   float64
6  Pedigree     768 non-null   float64
7  Age         768 non-null   int64
8  Outcome      768 non-null   int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```

Univariate analysis was performed to understand the distribution and characteristics of individual variables in the dataset. This involved visualizing each feature using histograms and density plots to identify patterns such as skewness, spread, and potential outliers. This step helps in understanding the behavior of each variable independently before exploring relationships between variables.

Univariate Analysis

Select a Feature for Analysis

Glucose



Key Insights

- Most variables show skewed distributions rather than normal distribution
- Features like Glucose and BMI exhibit wide variability, indicating strong predictive potential
- Some variables such as Insulin contain extreme values, suggesting possible outliers
- Age distribution indicates more observations among younger individuals
- Non-normal distributions suggest the need for scaling and careful model selection

Bivariate analysis was conducted to explore relationships between pairs of variables, particularly between predictor variables and the target variable (Outcome). This involved visualizing patterns using scatter plots and bar charts to identify how features such as Glucose, BMI, and Age influence the likelihood of diabetes. This step helps uncover trends, correlations, and potential predictors of the target variable.

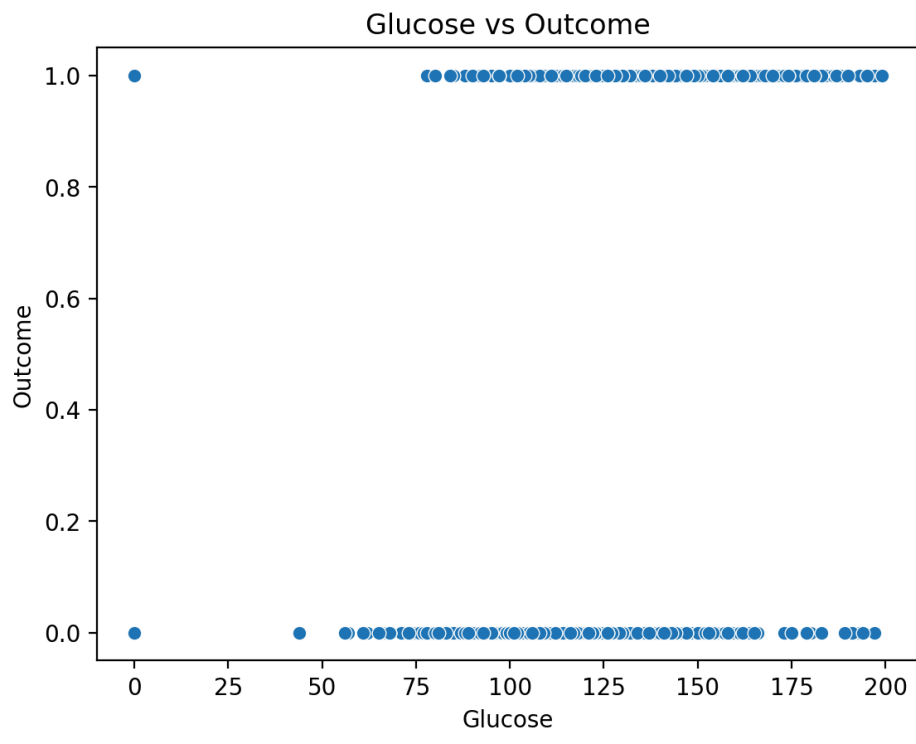
Bivariate Analysis

Select Feature (X)

Glucose

Select Target (Y)

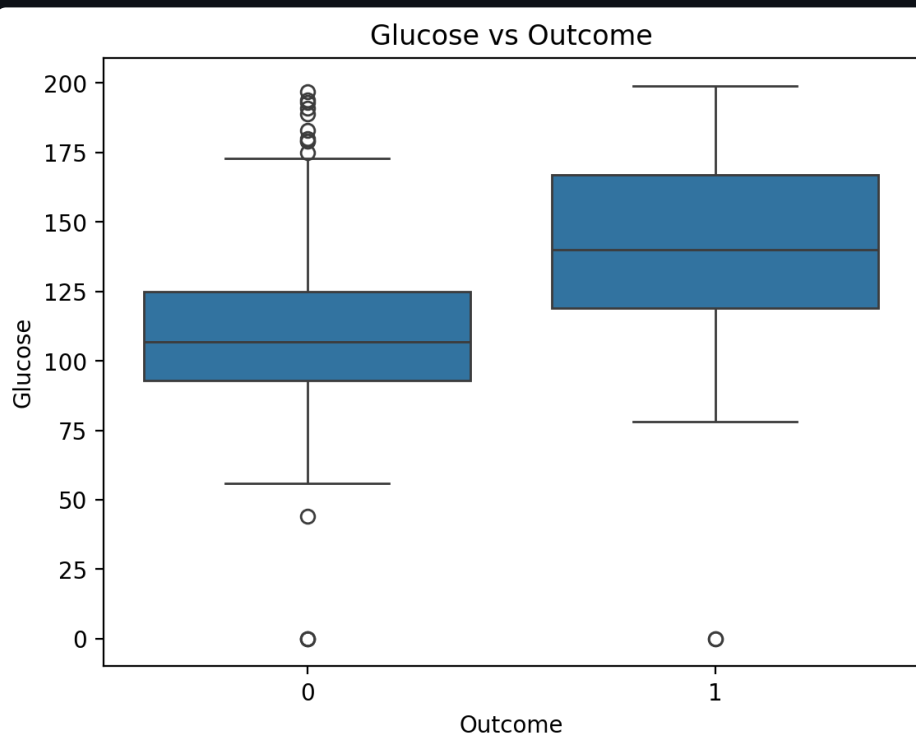
Outcome



Feature vs Outcome - Insights using boxplots grouped by Outcome:

Select Feature

Glucose



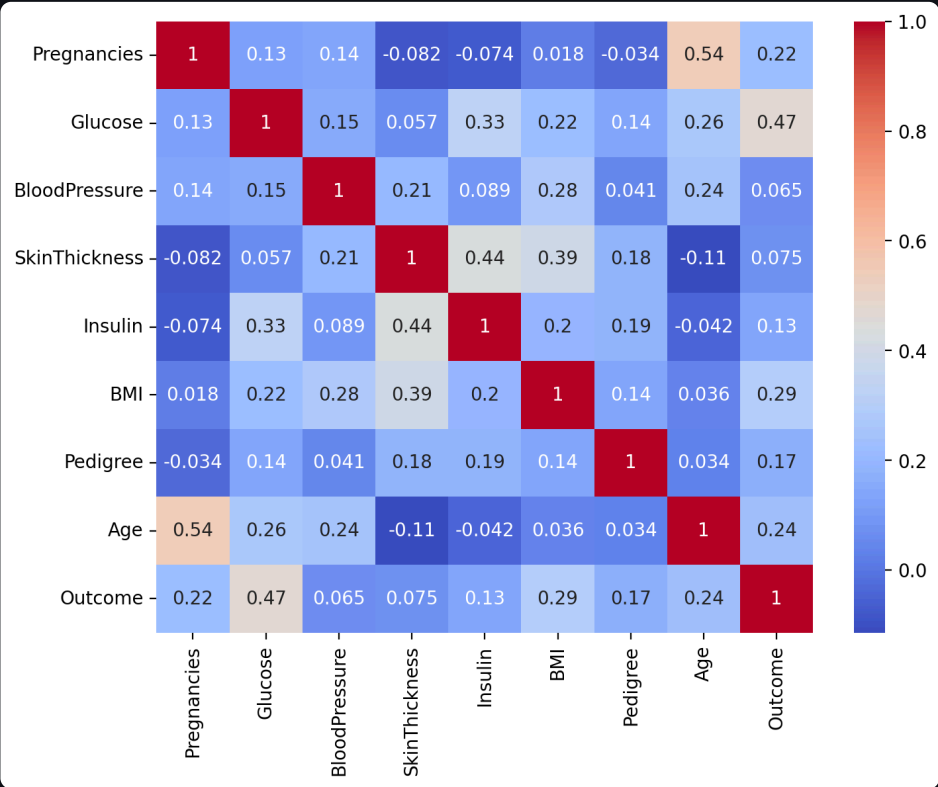
Key Insights

- Glucose shows a strong positive relationship with diabetes outcome
- Individuals with higher BMI are more likely to have diabetes
- Age demonstrates a moderate positive association with the target variable
- Some features such as Insulin show high variability with weaker predictive patterns

- lear separation in certain variables indicates their usefulness for classification models

Multivariate Analysis (Correlation Matrix - Correlation Heatmap)

Multivariate analysis was performed to examine relationships among multiple variables simultaneously. This was achieved using a correlation heatmap, which highlights the strength and direction of relationships between features. This step helps identify highly correlated variables, understand feature interactions, and determine which predictors have the strongest relationship with the target variable (Outcome).



Key Insights

- Glucose exhibits the strongest positive correlation with diabetes outcome, confirming it as the most important predictor
- BMI shows a moderate positive relationship, indicating body weight influences diabetes risk
- Age also demonstrates a moderate association with the target variable
- Most other features show weak correlations, suggesting limited predictive power individually
- There is no strong multicollinearity among predictors, meaning features are relatively independent

In this stage, the dataset was prepared for modeling by addressing data quality issues, separating features and target variables, and splitting the data into training and testing sets. Standardization was then applied to ensure that all features are on a similar scale, which improves the performance of machine learning algorithms such as Logistic Regression and KNN.

Data Cleaning

Replaced 0 values with NaN

Missing values handled using mean imputation

Feature Selection

Features:

▼ [

```
0 : "Pregnancies"
1 : "Glucose"
2 : "BloodPressure"
3 : "SkinThickness"
4 : "Insulin"
5 : "BMI"
6 : "Pedigree"
7 : "Age"
```

]

Target: Outcome

Data split into training and testing sets

Data successfully standardized

Key Insights

- Certain variables contained invalid zero values, which were treated as missing data
- Missing values were handled using mean imputation, ensuring no data loss
- The dataset was split into training (80%) and testing (20%) setse
- Feature scaling was applied to ensure all variables are on the same scale
- Standardization is especially important for distance-based models like KNN

Modelling

In this stage, multiple machine learning models were developed to predict diabetes outcomes. The selected models include Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN). Each model was trained on the prepared dataset and evaluated using performance metrics such as accuracy and AUC. This step aims to identify the most effective model for predicting diabetes.

Model Training

Models trained successfully

Model Performance

```
{
  "Logistic Regression": {
    "Accuracy": 0.7662337662337663
    "AUC": 0.8236914600550964
  }
  "Random Forest": {
    "Accuracy": 0.7337662337662337
    "AUC": 0.81643709825528
  }
  "KNN (k=11)": {
    "Accuracy": 0.7207792207792207
    "AUC": 0.7932047750229568
  }
}
```

Key Insights

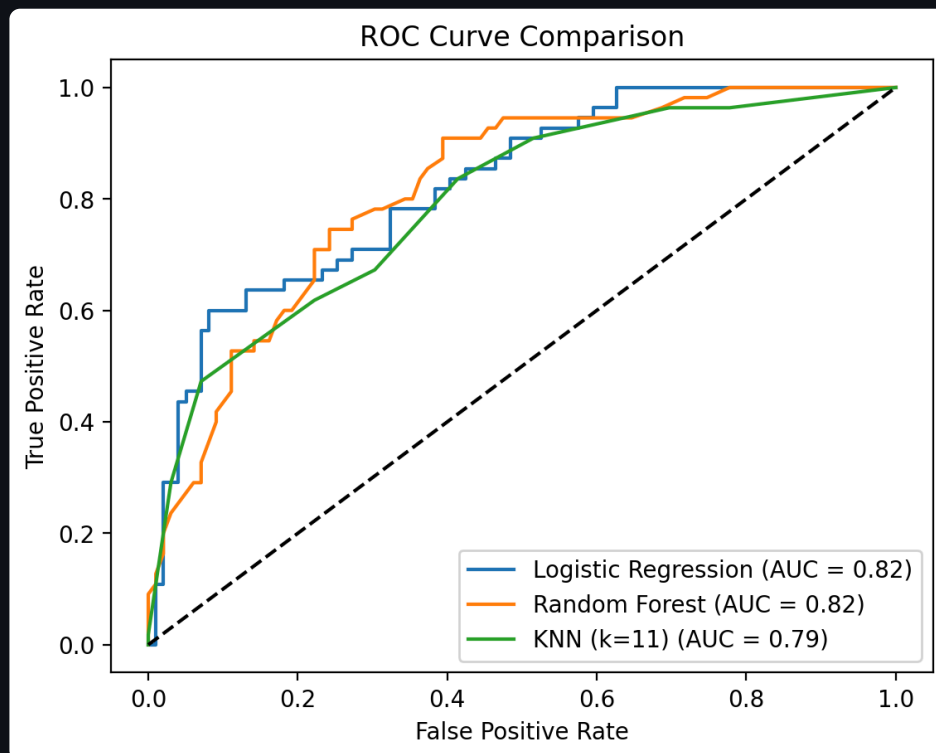
- Logistic Regression achieved the highest accuracy (76.6%), indicating that it made the most correct predictions among the models tested.
- Logistic Regression also recorded the highest AUC (0.824), showing that it has the best ability to distinguish between diabetic and non-diabetic cases.

- Random Forest performed slightly lower than Logistic Regression in both accuracy and AUC, suggesting that while it captures complex patterns, it may not generalize as well on this dataset.
- The KNN model ($k = 11$) showed the lowest performance, with accuracy and AUC of approximately 72% and 0.79 respectively, indicating that distance-based methods are less effective for this dataset.
- The close performance between Logistic Regression and Random Forest suggests that the dataset may have relatively strong linear relationships, which Logistic Regression captures effectively.
- All models achieved AUC values above 0.79, indicating that they have good classification capability and are significantly better than random guessing.
- Logistic Regression emerges as the best overall model, combining both high accuracy and strong AUC, making it the most reliable choice for predicting diabetes in this dataset.

Model Evaluation

The ROC (Receiver Operating Characteristic) curve was used to evaluate the classification performance of the models. It illustrates the trade-off between the True Positive Rate (sensitivity) and False Positive Rate across different thresholds. The Area Under the Curve (AUC) provides a single performance metric, where higher values indicate better classification ability. This visualization helps compare models beyond accuracy and confirms which model best separates the classes.

4. Model Evaluation (ROC Curve)



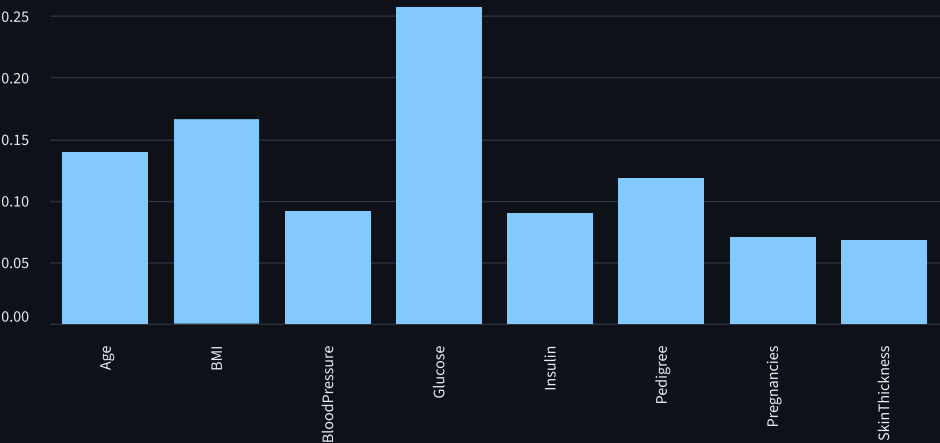
Key Insights

- The ROC curves show that Logistic Regression consistently stays closer to the top-left corner, confirming its superior ability to distinguish between diabetic and non-diabetic cases.
- Random Forest performs closely to Logistic Regression but falls slightly below it, indicating good but not optimal classification performance.
- The KNN model has a lower curve compared to the other models, reinforcing its relatively weaker ability to separate the classes.
- All models perform better than the random baseline, as their curves lie above the diagonal line, confirming that they have learned meaningful patterns from the data.

- The ROC analysis supports earlier findings from accuracy and AUC, validating Logistic Regression as the most effective model for this dataset.

Feature importance analysis was conducted using the Random Forest model to identify which variables contribute most to predicting diabetes. This helps in understanding the key factors influencing the outcome and enhances interpretability of the model.

5. Feature Importance (Random Forest)



Key Insights

- Glucose emerges as the most important feature, confirming its strong relationship with diabetes outcomes observed during EDA.
- BMI and Age also show significant importance, indicating their influence on diabetes risk
- Features such as BloodPressure and SkinThickness contribute less, suggesting limited predictive power
- The model relies on a small number of dominant predictors, reinforcing earlier correlation findings
- Feature importance aligns strongly with results from both bivariate and multivariate analysis, validating the overall model consistency

The study successfully applied the CRISP-DM methodology to analyze and predict diabetes outcomes using a combination of exploratory data analysis and machine learning models. Key insights were derived from the dataset, and model performance was evaluated using accuracy and AUC metrics.